

Transforming a general LP to standard form

Original problem

Let the general linear program be

$$\begin{aligned} & \text{minimize} && c^\top x + d \\ & \text{subject to} && Gx \preceq h \\ & && Ax = b \end{aligned}$$

where $G \in \mathbb{R}^{m \times n}$, $A \in \mathbb{R}^{p \times n}$, $c \in \mathbb{R}^n$, $d \in \mathbb{R}$, $h \in \mathbb{R}^m$, and $b \in \mathbb{R}^p$.

Step 1: Replace unrestricted variables by nonnegative pairs

We want to transform the problem to a form where all variables are positive or zero. For each component x_j introduce nonnegative variables $x_j^+ \geq 0$ and $x_j^- \geq 0$ and set

$$x_j = x_j^+ - x_j^-.$$

In vector form we define $x^+, x^- \in \mathbb{R}^n$ and thus write

$$x = x^+ - x^-, \quad x^+ \geq 0, \quad x^- \geq 0.$$

Step 2: Convert inequalities to equalities introducing slack variables

Introduce slack variables $s \in \mathbb{R}^m$ with $s \geq 0$ to convert $Gx \preceq h$ to an equality:

$$G(x^+ - x^-) + s = h, \quad s \geq 0.$$

Step 3: Stack variables and build standard-form system

Form the stacked decision vector

$$x' = \begin{bmatrix} x^+ \\ x^- \\ s \end{bmatrix} \in \mathbb{R}^{2n+m}, \quad x' \geq 0.$$

Define the block matrix B and right-hand side e by

$$B = \begin{bmatrix} G & -G & I_m \\ A & -A & 0_{p \times m} \end{bmatrix}, \quad e = \begin{bmatrix} h \\ b \end{bmatrix}.$$

Then the constraints become the single equality

$$Bx' = e.$$

Objective function in the new variables

The original objective transforms as

$$c^\top x + d = c^\top (x^+ - x^-) + d = p^\top x' + d, \quad p = \begin{bmatrix} c \\ -c \\ 0_m \end{bmatrix}.$$

where 0_m are m zeros in the vector. The constant d does not affect the minimization, so we can drop it.

Standard-form LP

Discarding the additive constant d in the optimization problem (it may be added back to report the original objective value), the equivalent standard-form LP is

$$\begin{aligned} & \text{minimize} && p^\top x' \\ & \text{subject to} && Bx' = e \\ & && x' \succeq 0 \end{aligned}$$

Relation between feasible sets, optimal solutions, and optimal values

Feasible sets. Let

$$F = \{x \in \mathbb{R}^n \mid Gx \preceq h, Ax = b\}$$

be the feasible set of the original LP and

$$F' = \{x' \in \mathbb{R}_+^{2n+m} \mid Bx' = e\}$$

the feasible set of the standard-form LP. Define the canonical mapping

$$\Phi : x \mapsto x' = (x^+, x^-, s) = (\max(x, 0), \max(-x, 0), h - Gx).$$

Then Φ maps F into F' . With the canonical choice $x^+ = \max(x, 0)$ and $x^- = \max(-x, 0)$ the map is bijective onto F' and invertible by $x = x^+ - x^-$. Thus feasibility is preserved exactly.

Optimal solutions. If x^* is optimal for the original LP then $x'^* = \Phi(x^*)$ is feasible for the standard form and attains

$$p^\top x'^* = c^\top x^*.$$

Conversely any optimal $x'^* \in F'$ yields $x^* = x'^* - x'^{-*} \in F$ with the same objective value (up to the constant d). Nonuniqueness in the split (x^+, x^-) can produce multiple x' corresponding to the same x if one does not fix the canonical split. Using the canonical split removes that ambiguity.

Optimal values. Let v_{orig} denote the optimal value of the original LP and v_{std} the optimal value of the standard-form LP with objective $p^\top x'$. Then

$$v_{\text{orig}} = v_{\text{std}} + d.$$

Therefore we can recover the optimal value easily.